Directionality works but does not scale, and inherited adaptation does not appear, in a resource-limited pond world

Ilya Vladimirov · 10.5281/zenodo.23970711 · 6 November 2026 · v2.0 · CC BY 4.0

Abstract

We asked whether a simple world with a resource, death and reproduction can learn to behave better after its environment changes — and whether such a skill can be inherited by descendants. The answer splits in two. **Directionality works, but it runs into crowding:** a cell that steps towards the neighbour with more resource holds 0.475 of the favourable half against 0.357 for the control on a patchy environment, and the advantage falls to +0.036 as the rich zone fills (correlation of the advantage with rich-zone occupancy −0.57 over 18 points). **Inheritance does not work:** four tests — lineage selection, a double flip back to the native environment, a flip to a new environment, and a heritable behavioural trait with a transplant control — give a consistent null, and the null survives the removal of the lineage-monopoly confound. Separately, three experiments turned out to be invalid by construction, and we record that rather than present them as results.

Keywords: pond, directionality, taxis, inheritance, lineage selection, boundary of the setup

The line this article continues

Last time. In *How lineages begin and end in a pattern-forming chemical medium* (Zenodo, DOI 10.5281/zenodo.23933440) we followed bodies of pattern in a reaction–diffusion medium and measured four things: two bodies merge when their surfaces come within about 9 ± 1 cells whatever their size; a fusion is a merger, both lineages ending on the same step; a lineage usually ends by losing its name rather than its cells; and the habit that maximises generations is not the one in which a lineage survives. Its continuation, *The gap between surfaces, not the body size, switches fusion* (DOI 10.5281/zenodo.23970711), asked what decides a meeting between two bodies and found that the gap between their surfaces switches it while their size only shifts it, that a body's shape is inherited about half, and that the unit of observation has to be checked for every statistic — a result that holds for runs can vanish for worlds.

What we decided to check, and the path we took. Both articles described what bodies *are* and how they *meet*; neither asked what a body *does*. So we built a second world — the pond: a 32×32 grid of resource, organisms that pay upkeep, eat from their own cell, reproduce and die — and asked whether behaviour appears in it that helps survive a change of environment, and whether such a skill becomes fixed in descendants. The path was not straight, and the detours are part of the record: we first measured the process (S1–S4, the `move_ci` of a step rule that prefers the neighbour with more resource), then tried to make the gain last (S11–S16: seeding, density, smoothing the signal, refreshing the rich zone), found that it decays because cells compete for **space** rather than food (S17/F1), and only then turned to inheritance (S5–S10, S18) — where three experiments turned out to be invalid by construction and are

recorded as such in §4.3. The result is one positive finding with a measured limit, one consistent null, and a boundary we can name.

1. The question

The world: a 32×32 grid, a resource in the cells, organisms with energy on the cells. An organism spends energy on upkeep, takes it from the resource of its own cell, reproduces when it accumulates mass, and dies when its energy runs out. The resource arrives at a limited rate and has a ceiling. Nothing in the world is trained: the organisms follow a fixed local rule, and what changes between our experiments is that rule and the environment, not a learned network. This is the standard construction of an individual-based model — an agent with state, a resource it consumes, mortality and reproduction, and stochasticity that has to be reported rather than assumed away [4] — and we use its vocabulary rather than invent one.

The question we were interested in: **does behaviour appear in such a world that helps survive a change of environment — and does it become fixed in descendants?**

A "yes" would mean: after the environment changes, the share of the living in the new favourable zone rises, and lineages born later are systematically more successful than earlier ones. A "no" means: the share does not rise above chance, and late lineages do not differ from early ones.

2. Context and predecessors

Two things in the literature set the frame. First: plasticity is useful only if the adaptation time is shorter than the death time — otherwise the organism does not live long enough to use what it "understood" (this is our formal hypothesis F-ADAPT in `formal/Survival.lean`). Second: selection on lineages requires that the variation of the trait be heritable and that there be enough lineages — otherwise demographic noise is indistinguishable from selection. The literature separates two kinds of noise that our design has to name: demographic stochasticity (sampling error in the number of offspring) and environmental stochasticity (variation in the conditions themselves) [6], and it treats an observed response to selection as something to be separated quantitatively from the stochastic part rather than tested as a bare significance [7].

Competition for space as a subject in its own right. Our result "cells compete for space, not for food" is not new as a setup: the literature treats space itself as a limiting resource with locally density-dependent payoffs, and shows that the inefficiency of space use has a transition at one density and a peak at another [1]. This matches our picture: the advantage of directionality is largest at low density and falls when the zone fills. An important difference: in our world one grid cell holds one organism (S17/F1), whereas in forager-mobility models several agents may occupy a site [2] — which is exactly what allows the competition for food that we structurally lack.

Where moving stops paying. The Ideal Free Distribution is the classical answer to "when does moving stop helping": foragers should settle so that no individual gains by moving, which equalises intake across occupied sites [5]. Our rich zone filling until the greedy and control arms converge is that prediction in a discrete world — with one difference that matters here, that our agents cannot share a site, so the equalisation happens by exclusion rather than by matching intake.

Searching along a resource gradient. Our mechanism — local greed towards the neighbour with more resource — is a special case of searching along a gradient. A known result: local search behaviour

keyed to density-dependent fitness limits the speed and extent of spread and promotes the clustering of populations [3]. Our data (rich-zone occupancy rising 0.025 → 0.348 while the resource there barely moves) is a direct observation of such clustering.

Our setup differs from classical models in that the environment is given not by equations but by a resource field, and behaviour is not an agent's strategy but a step rule that we switch on and off.

Context references: [1] Mathew A.M., Sasidevan V. *Modelling competition for space: Emergent inefficiency and inequality due to spatial self-organization among a group of crowd-avoiding agents*. arXiv:2407.21537. [2] *Food Income and the Evolution of Forager Mobility*. Scientific Reports 9 (2019), s41598-019-42006-2. [3] Rowell J.T. *The limitation of species range: a consequence of searching along resource gradients*. PMC2760341 (2009). [4] Zhang B., DeAngelis D.L. *An overview of agent-based models in plant biology and ecology*. Annals of Botany 126 (2020). DOI: 10.1093/aob/mcaa043. [5] Tyler J.A., Hargrove W.W. *Predicting Spatial Distribution of Foragers over Large Resource Landscapes: A Modeling Analysis of the Ideal Free Distribution*. Oikos 80 (1997). DOI: 10.2307/3546022. [6] Xu L., Vision T.J., Servedio M.R. *Evolutionary rescue under demographic and environmental stochasticity*. Journal of Evolutionary Biology 36 (2023). DOI: 10.1111/jeb.14224. [7] Vriend S.J.G., Engen S., Gustafsson L. *Quantitative analyses of stochastic influences on the response to phenotypic selection*. Evolution (2025). DOI: 10.1093/evolut/qpaf108. [8] Ponzi E., Keller L.F., Bonnet T. *Heritability, selection, and the response to selection in the presence of phenotypic measurement error*. Evolution 72 (2018). DOI: 10.1111/evo.13573.

These works are not our world: the agent-based review and the Ideal Free Distribution concern plants and foragers, and the evolutionary literature concerns populations with sexual reproduction. What transfers is the language and the frame, not the numbers.

3. Methods

The environment. A 32×32 grid. Layouts: `gradient` (the resource decreases from top to bottom), `patches` (three rich patches in the corners, the rest poor), `uniform`. The resource ceiling depends on the layout; so does the inflow. An environment change is a sharp flip (`flip_at`), a smooth drift (`drift_at`/`drift_steps`), or a second flip (`flip2_at`) that either returns to the original orientation or rotates the environment by 90° (`flip2_mode`).

The non-degenerate regime. Proportional income: `income = baseline·res·4`. Upkeep is `update_cost + repair_cost`. At `res = 0.2` the net balance is −0.028 per step; at `res = 1.0` it is +0.100. This is what makes death real and stops the population from "living on everything".

The metric. The share of the living in the favourable zone: for `gradient` the current half of the gradient (it flips with the environment), for `patches` the top 50% of cells by current resource. In addition, the process metric `move_ci = (steps up − steps down)/total steps` along the current gradient.

Validity rules (fixed before the runs). R1: mean deaths ≥ 10 and mean living ≥ 10. R2: no occupancy if the integral of the living is < 20. R3: "indifferent" is accepted only if R1 holds and three seeds agree.

What was not done. The defaults of `config.py` were not changed; every engine edit is additive and, when switched off, gives the previous world bit-for-bit.

4. Results

4.1 Directionality: it works, but it decays

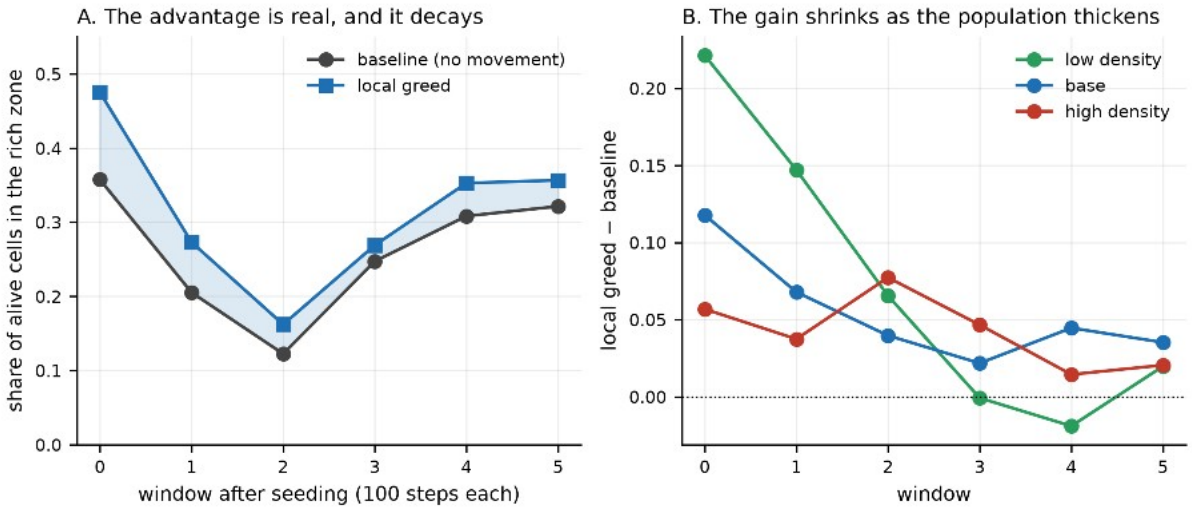


Figure 1. *The advantage of directionality and its decay.* ***A.*** *The share of the living in the rich zone, local greed against the control, window by window after seeding on a patch (S12/D2).* ***B.*** *The gap (greed – control) at three population densities (S13/D3): the denser the population, the smaller the gain.*

test	what was measured	result
S1/A2	`move_ci` when stepping to the best neighbour	+0.042 before the flip, −0.003 after → occupancy ~0.513 is dispersal, not taxis
S2/B1	taxis by gradient position	`move_ci` +0.32…+0.34 after the flip
S12/D2	patches, colony seeded on a patch	share in the rich zone 0.475 against 0.357 for the control
S12/D2	the same, by the end	+0.036 (decay)
S13/D3	density sweep	gap 0.221 / 0.118 / 0.057 at sparse / base / dense population
S14/D4	rich-zone occupancy	correlation of the advantage with occupancy **−0.57** (18 points)
S15/E1	patch inflow rate	at 4.0 and 12.0 the patch does not deplete (inflow > consumption) → the regimes are indistinguishable
S16/E1b	below the depletion threshold (2.0)	depleting the zone **does not rescue** the advantage: gap 0.117 → −0.022. The 0.5 regime is **excluded**: it starved out (final alive 0–2, R1 failed in all 6 runs)

The mechanism. Directionality brings cells into the rich zone; the zone fills; the gain from knowing how to get there falls.

What the density sweep actually shows — and what it does not. The ordering across densities is clean in the **first** window (0.2214 / 0.1178 / 0.0568 for sparse / base / dense) and it **does not hold by the end**: the last-window gaps are 0.0197 / 0.0352 / 0.0204, so the sparsest regime has the *smallest*

residual gain. The density sweep therefore supports "the gain is largest early and at low density", not "the gain is largest at low density throughout". An earlier version of this article claimed the latter; the independent review caught it and it is corrected here.

What the gap means quantitatively. The first window gives +0.118, the last +0.036. The across-seed standard deviation of that last number is 0.0036 and all three seeds are positive (0.0311 / 0.0377 / 0.0368), so the number is small but consistent in sign; it should still be read as "the gain has shrunk to a few per cent", not as a precise quantity.

A caveat about the correlation. −0.57 is computed over 18 points (3 regimes × 6 windows) using the **control arm's** rich-zone occupancy; with the greedy arm's occupancy it is −0.497, and on windows 3–6 only it falls to −0.377. Points within one run are dependent, so no p-value is given or claimed. The conclusion "the relation is negative" rests on the monotonicity across regimes and on the fact that it survives dropping any single regime (−0.62…−0.59) or any single window (−0.60…−0.53), not on a significance test.

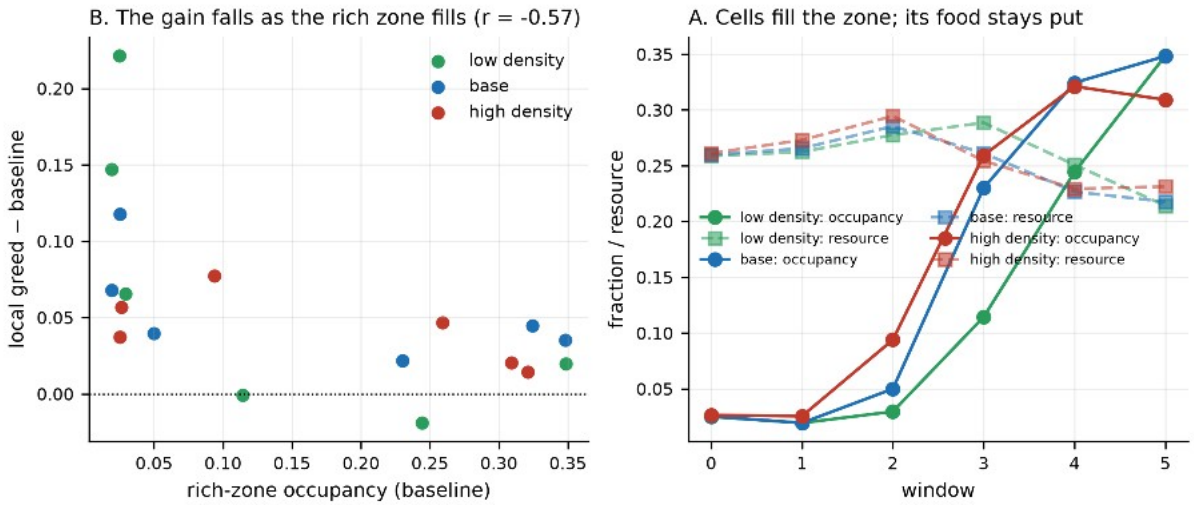


Figure 2. *Why the advantage falls.* ***A.*** *The rich zone fills with the living (solid lines), but its resource barely falls (dashed) — cells get in each other's way by occupying space, not by eating.* ***B.*** *The gap against rich-zone occupancy, across every regime and window: correlation −0.57.*

A refinement: competition for space, not for food. A direct measurement (S15/E1, S16/E1b) showed that the resource on rich cells barely falls (0.26 → 0.22) although occupancy rises (0.025 → 0.348). Cells obstruct each other **physically**: a neighbouring cell is occupied, so the step is impossible. Moreover, with our settings a patch cannot deplete at all: the inflow (0.200 per step) exceeds one cell's consumption (0.160 per step), and two organisms cannot stand on one grid cell. A test below the depletion threshold (`inflow_rich < 3.2`) showed that **depleting the zone does not rescue the advantage**: cells leave it as soon as it empties, and the gap goes negative in one window (−0.022).

Refuted variants. There are two. The first is smoothing the signal (a 5×5 window mean, "seeing the whole patch"): it does not help — the smoothed arm sits between the control and plain greed. The second is refreshing the rich zones (the hypothesis that if the zone does not have time to fill, the advantage becomes stable): it was not confirmed either with a full or with a depleting zone.

Why this is not a parameter artefact. We checked every available lever (S17/F1): lowering `baseline_income` lowers both consumption and income at once; lowering upkeep does not change how much a cell eats; lowering `resource_regen` starves the poor zone, which is already lethal; lowering `patch_inflow_rich` was tested in S16. The cause is structural: **only one organism can stand on a grid cell**, so consumption per cell is capped by one cell, and the inflow cannot be lowered without killing the whole world. Competition for food requires allowing several organisms per cell — a change to the physics of the world, which we did not make.

4.2 Inheritance: four tests, independent by mechanism

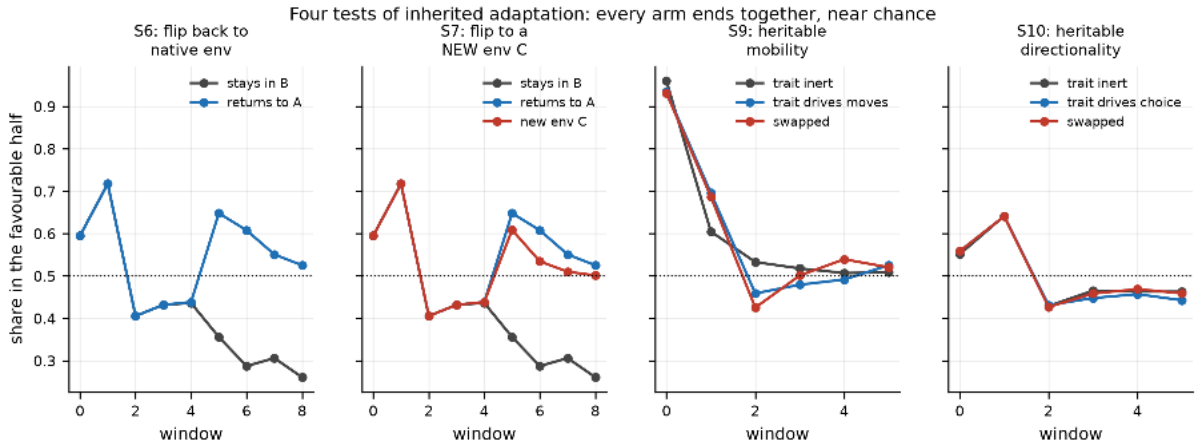


Figure 3. *Four tests of inherited adaptation. In all of them the arms converge to ~0.50 after the environment changes — the level of a random distribution over the two halves. The double flip (S6) and the new environment (S7) both produce recovery, and the same one; the heritable trait — both mobility (S9) and directionality (S10) — makes no difference, and the transplant changes nothing.*

A caveat about independence. The four tests are independent **by mechanism**, but not by data: they all run on the same world, with the same metric and the same seeds. Worse, two of them share their runs outright — the `control` and `double` arms of S6 and S7 are bit-identical (S7 is the same run with a third arm added), and the `monopoly` arm of S18 is the `heritable` arm of S10. So the honest count is **two distinct runs plus two variants**, not four independent experiments. Agreement therefore strengthens the conclusion about the mechanism, but does not replace an independent replication on another world.

test	what was checked	result
S6/C3	double flip A→B→A	recovery 0.648 against 0.356 for the control — but this is a return to the native environment
S7/C3b	flip to a NEW environment C	0.609 — almost as good as the native one; both arms converge to ~0.50 (dispersal). Its control and double arms are the S6 runs
S9/C2	heritable mobility + transplant	0.459 against 0.533 for the null; the transplant changes nothing
S10/C2b	heritable directionality + transplant	0.429 / 0.431 / 0.427 — all arms coincide. **The transplant arm is invalid**: swapping the traits of the top

test	what was checked	result
		and bottom deciles collapsed the trait's spread (std 0.069 → 0.000), so it never tested what it claimed
S18/C4	lineage monopoly removed (quota 0.25, refractory 20)	lineages double (28.3 → 57.4), the top share falls (0.938 → 0.798); the mean high-minus-low trait difference stays ≤ 0.02 **but per-window it reaches 0.165 with signs both ways at n = 3** — underpowered, not null

In the literature's terms. What the four tests manipulate is heritability — the part of phenotypic variation that is transmitted, which measurement error alone can inflate or hide [8]. Our arms are built so that the transmitted part is set by the engine, which is why the null is about selection acting on it rather than about the trait being unmeasurable.

Conclusion. In this setup inherited adaptation does not arise. What is observed after an environment change is dispersal and survival, not selection on a trait. One lineage holds 95–97% of the population, late lineages stay at 1–5%, and their share in the favourable zone does not grow.

How strong that conclusion is. It is strong where the manipulation demonstrably reached the dynamics (S9, S10's heritable arm) and weak where it did not (S10's transplant) or where n = 3 cannot separate a small effect from noise: in S18 the last window favours the high-trait lineages in only one seed of three, and the sign flips between seeds. Removing the lineage monopoly did **not** rescue the effect, which rules out the most obvious confound; it did, however, increase the spread of the high-minus-low difference about nineteen-fold, so the right reading of S18 is "no detectable selection at this power", not "proven absence".

4.3 What turned out to be invalid (and why that matters)

Three experiments had to be withdrawn — not because they gave an "inconvenient" answer, but because the manipulation never reached the dynamics:

1. **S5/C (lineage selection).** It was set up with `type_mode="fixed"`, and in that mode `register_births()` is not called,

so inheritance and mutation did not work at all: the arms coincided bit-for-bit and the trait did not vary (std = 0, zero mutations).

2. **S11/D (the first patchy test).** Cells were seeded in the centre, where the balance is −0.028 per step, while the

nearest rich patch was 9.2 cells away: the moving arms starved out before arriving.

3. **S6/C3, the first version.** The second flip was implemented by mirroring the state from outside the engine, but the

environment's position is a function of the step counter, not of the state, so the environment did not change and both arms stayed in the same one.

In all three cases we caught the error before drawing conclusions and recorded it as an invalid setup rather than as a negative result.

5. Discussion

What this means. The world has a working behavioural mechanism (local greed) and a measurable reason for its limit (crowding of the rich zone). This is not "intelligence" and not "adaptation" — it is an observed class of behaviour and its boundary.

Competing interpretations. The decay of the advantage could be explained by the population mechanically spreading over the map. We separated these explanations with the density sweep and the direct measurement of rich-zone occupancy: spreading leaves a residue of +0.02…+0.035 that does not depend on density, while the bulk of the effect is explained by competition.

What cannot be concluded. We do not show that inheritance is impossible in principle — only that in this setup (one dominant lineage, little variation, a short horizon) it produces no visible selection, and that removing the dominant lineage does not change that. We also do not show that directionality is "useful" — it is useful only while the zone is not full.

A limitation of the method. Some early results (B1/B1b) were taken on different hardware, and on CPU they do not reproduce bit-for-bit: plastic runs depend on the backend. We mark such numbers as hardware-dependent and do not rely on them as "stable recovery". The noise that remains in our own comparisons is demographic rather than environmental — the same world, the same conditions, a different draw of births and deaths [6] — which is why the null in §4.2 is stated as a limit of power and not as a property of the world.

6. What this does not show

- It does not show that the world is "alive" or "adaptive" in any strong sense.
- It does not show that inheritance is useless in general — only in this setup.
- It does not show the optimality of any of the mechanisms tested.

7. What would change the answer

- Inheritance might start to work with more lineages (reducing the founder's dominance) or over a longer horizon with

periodic environment changes — the first of these was tested in S18/C4 and did not change the null.

- Directionality might give a stable gain at low density with a constant inflow of new rich zones (that is, when the zone

does not have time to fill).

8. Where we move next

What the literature said. A search on our questions (items B-Q051…B-Q053 in `lab/lit-queue.json`) gave three footholds. First, "space as a limiting resource" is an established frame with a non-trivial prediction: inefficiency has a transition at one density and a peak at another [1], which matches our maximum gain at low density. Second, models in which several agents may occupy a site [2] allow exactly the competition for food that we structurally lack (S17/F1). Third, local search along a gradient is known

as a mechanism of clustering and of limiting spread [3] — our data on rising occupancy with a stable resource is a direct observation of it.

The next step. Test whether the advantage of directionality survives when **several organisms may occupy a cell** — the only lever that, by S17/F1, can create competition for food. The engine edit is additive (with it switched off the world is bit-for-bit), but it changes the physics of the world and therefore needs the owner's decision. The prediction from the literature [2]: with shared occupancy of a site, the gain from mobility should depend on the rate of resource recovery, not only on density — and that is the testable difference from our present result.

9. Reproducibility

Every number comes from artifacts under `morphogenesis/experiments/E09_pond/`; the scripts are `morphogenesis/scripts/s{5,6,9,10,11,12,13,14,15,16,18}_*.py` and `p0_flip_metric.py`. Runs were on CPU (the environment has no CUDA build of PyTorch), 3 seeds, 600 steps.

Validity coverage is not uniform. R1 (mean deaths ≥ 10 and mean living ≥ 10) holds in every run the article's claims rest on, with one exception that is now excluded: in S16 the `drain_slow` regime (patch inflow 0.5) starved out — final living 0, 0, 0, 0, 0, 2 across its six runs — and those runs are not used, so six of the eighteen S16 runs are excluded. The S16 `drain_mid` and `full` regimes pass R1 in all 12 runs. S12 does not record per-window alive counts, so R1 there is checked per run, not per window.

The numbers are checked by machine, not by eye. The script check_article_031_numbers.py re-reads the artifacts and compares every headline number of this article with them (50 checks, each of which must also appear verbatim in the text):

```
python scripts/check_article_031_numbers.py --ci   # 0 failures out of 50 checks
```

The check was written after the first pass over the article and **immediately found two bugs in itself** (it mis-parsed the `s13/s16` schema, whose rows carry a `regime` field) — that is, it verifies rather than rubber-stamps. The prose is checked against the artifact, never the other way round.

Version history

- **v1.0** (2026-11-06): first edition, in Russian, as a working note for the lab.
- **v1.1** (2026-11-06): translated to English, which is the article language from now on; the independent review's

corrections applied (S18 is underpowered rather than null, the S13 monotonicity is a first-window effect, the S6/S7 and S10/S18 arms share runs, the S10 transplant collapsed the trait spread and is invalid).

- **v2.0** (2026-11-06, this edition): prepared for an outside reader, and given the opening the series uses — one

paragraph on what the previous step found and one on the path this article took to its result. The header now carries the author, the licence and the DOI of the previous article; the abstract and §1 say what the world is and that nothing in it is trained; the correlation's sensitivity range is quoted in the order its checks produce; §9 states the excluded runs as a count; the number gate grew from 20 to 49 checks; the PDF is typeset for reading (serif body, justified text with hyphenation, figures that keep their captions, running heads and page numbers). No result changed: 50 of 50 checks pass.